Exercise 40. This exercise shows how to turn around the arrows in the previous two exercises. Let D be a category and let $G\colon D \to \mathbf{Set}$ be a (covariant!) functor. Define $\int G$ to be the category whose objects are elements of G and whose morphisms $(c, x) \to (d, y)$ are the morphisms $f\colon c \to d$ in C such that $Gf(x) = y$. This category is also denoted $\int G$, although this notation can lead to confusion, see item a.

a. Let $C := D^{\mathrm{op}}$ and let $G'\colon C^{\mathrm{op}} \to \mathbf{Set}$ be the same functor as G, but viewed contravariantly. Show that the category $\int G$, as defined in this exercise, is the same as $(\int G')^{\mathrm{op}}$, as defined in Definition G.

b. Write $\mathbf{y}'\colon D^{\mathrm{op}} \to \mathbf{Set}^D$ for the Yoneda embedding of D^{op} and, for any object d of D, $\mathsf{e}'_{c,G}$ for the bijection $\mathbf{Set}^D(\mathbf{y}'c, G) \xrightarrow{\cong} Gc$. An element (c, x) of G is *universal* if the unique natural transformation $\alpha\colon \mathbf{y}'c \Rightarrow G$ with $\mathsf{e}'_{c,G}(\alpha) = x$ is an isomorphism. Prove that an element (c, x) of G is universal if, and only if, (c, x) is an initial object in $\int G$.

Conclusion: the meaning of 'universal element' depends on whether the functor is seen as covariant or contravariant.

a. The definitions of $\int G$ (ex. 40) and $\int G'$ (ex. 38) are the same, up to a change of direction.

b. $y : C \to Set^{C^{op}}$ $\qquad e'_{c,G} : Set^D(y'c, G) \xrightarrow{\cong} Gc$

$y' : D^{op} \to Set^D$

By ex. 39, (c,x) is universal $\Leftrightarrow$ (c,x) is terminal in $\int G'$ where G' is the contravariant functor of (a).

$\Leftrightarrow$ (c,x) is initial in $(\int G)^{op} = \int G$

↳ for G covariant.

Definition I. A *universal arrow* from an object c of C to a functor $U\colon D \to C$ is a pair (d, u), where d is an object of D and $u\colon c \to Ud$ is a morphism such that, for any morphism $f\colon c \to Ue$, there exists a unique morphism $f'\colon d \to e$ in D such that $Uf' \circ u = f$.

Exercise 41. Let $U\colon D \to C$ be a functor, let c be an object of C and consider the functor $H\colon D \to \mathbf{Set}$ given by composing U with $\mathbf{y}'c\colon C \to \mathbf{Set}$. Prove that a universal arrow is the same thing as a universal element of the covariant functor H.

Recall: left adjoint of U can be given by specifying, for each object c of C, a universal arrow from c to U.
(Then you can get $F: C \to D$ and a natural iso $D(Fc, d) \cong C(c, Ud)$.)

$H: D \xrightarrow{U} C \xrightarrow{y'c} \mathbf{Set}$. We saw in ex. 40 that a universal element of H is the same thing as an initial object of $\int H$.

($yc(a) = C(a,c)$, $y'c(a) = C^{op}(a,c) = C(c,a)$)

What is $\int H$?

Objects: (e, f) where $f \in He = (y'c)(Ue) = C(c, Ue)$

i.e. e an object of D and $f: c \to Ue$ a morphism in C.

Morphisms: $(d,u) \xrightarrow{f'} (e,f)$ is a morphism $f': d \to e$ in D such that $(Hf')(u) = f$

What is Hf'?

d	Ud	$C(c, Ud)$
$f' \downarrow$	$\downarrow Uf'$	$\downarrow Uf' \circ -$
e	Ue	$C(c, Ue)$
in D	in C	in $\mathbf{Set}$

$\to Uf' \circ u$

To conclude:

(d,u) universal element of H
$\iff$
(d,u) is initial in $\int H_c$
$\iff$
(d,u) is universal arrow from c to U

12 Cartesian closed categories

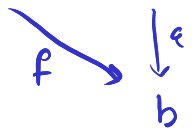
type in STLC ⇝ object in CCC
term ⇝ morphism in CCC
} up to βη

Recall that a *cartesian closed category* is defined as a category C with finite products such that, for any object a of C, the functor $(-) \times a \colon C \to C$ has a right adjoint; we will write $(-)^a \colon C \to C$ for this right adjoint.[3]

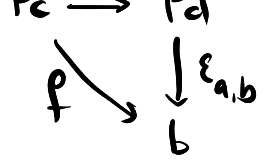
Exercise 42. Let C be a category with finite products. Prove that C is cartesian closed if, and only if, for any objects a, b of C, there exist an object b^a in C and a morphism $\epsilon_{a,b} \colon b^a \times a \to b$ such that, for any morphism $f \colon c \times a \to b$, there exists a unique morphism $f' \colon c \to b^a$ such that $\epsilon_{a,b} \circ (f' \times \mathrm{id}_a) = f$.

$\forall a, b \; \exists b^a \in \mathrm{ob}\, C, \; \epsilon_{a,b} : b^a \times a \to b, \quad \forall \; c \times a \xrightarrow{f} b \quad \exists! \; c \xrightarrow{f'} b^a \;$ s.t. $\; c \times a \xrightarrow{f' \times id_a} b^a \times a \quad (*)$

$C(a,b)$



let $a \in \mathrm{ob}\, C$.
Let's denote by $P : C \to C$ the functor $(-) \times a$.
P has a right adjoint $\Leftrightarrow$ for every object b of C, there exists a universal arrow from P to b.
$\Leftrightarrow \forall b \in \mathrm{ob}\, C$, there exists an object d of C and a morphism $Pd \xrightarrow{\varepsilon_{a,b}} b$ such that
for any $f : Pc \to b$, there exists a unique $f' : c \to d$ in C with



This is precisely condition $(*)$.

b^a is called "exponential" / "internal hom" $\qquad$ $\varepsilon_{a,b}$ is called "evaluation" / "co-unit".

[3] In the literature, you may also find the notations $[a \to b]$ or $a \Rightarrow b$ instead of b^a.

In Set: Given f, define $f' : C \to B^A$ as

$$f'(y) := \lambda x.\ f(y,x), \text{ where } y \in C, x \in A.$$

Indeed, given $(y,x) \in C \times A$, we have

$$\varepsilon(f' \times id_A(y,x)) = \varepsilon(f'(y),x) = f'(y)(x) = f(y,x),$$

and f' is the unique such function.

$B^A := \{f : A \to B \mid f \text{ a function}\}$

$\varepsilon : B^A \times A \to B$

$(f,x) \longmapsto f(x)$

Exercise 43. Let P be a partially ordered set. Characterize in order-theoretic terms what it means for the category associated to P to be cartesian closed.

P has finite products $\iff$ P has finite infima. $\iff$ P has a top element and for any $a, b \in P$, $a \wedge b = \inf(a,b)$ exists in P.



P has exponentials $\iff \forall a, b \in P$, there exists $b^a \in P$ with $b^a \wedge a \leq b$ such that for every $c \in P$ with $c \wedge a \leq b$, we have $c \leq b^a$.

$\iff \forall a, b \in P$, $\{c \in P \mid c \wedge a \leq b\}$ has a maximum, b^a (or $a \Rightarrow b$).

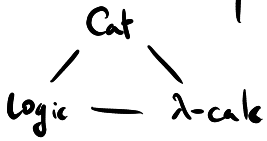
P has finite coproducts $\iff$ P has finite suprema (not implied by P being ccc, example?)

So P is a Heyting algebra $\iff$ P is a ccc with finite coproducts.

(for P a poset)

model of intuitionistic propositional logic

Research: how to extend this?



This gives a categorical explanation of the correspondence between IPC and STLC with finite sum types.

Exercise 44 (Exam 2023). Let C be a cartesian closed category in which finite coproducts exist. Construct, for any objects a, b, c, a morphism $c^a \times c^b \to c^{a+b}$.

First, logically, this means: $(a \to c) \wedge (b \to c) \vdash (a \vee b) \to c$.

with λ-terms, this means: $f_a : a \to c,\ f_b : b \to c \vdash ? : (a+b) \to c$

$[f_a, f_b]$

Since $(-) \times (a+b) \dashv (-)^{a+b}$, it's equivalent to construct $(c^a \times c^b) \times (a+b) \to c$.

Sine $d \times e \cong e \times d$, it's equivalent to construct $(a+b) \times (c^a \times c^b) \to c$

Using $(-) \times (c^a \times c^b) \dashv (-)^{c^a \times c^b}$ —"— —"— $a+b \to c^{(c^a \times c^b)}$ $d := c^a \times c^b$

By the universal property of coproduct, —"— —"— two arrows, $a \to c^d$ and $b \to c^d$.

Using $(-) \times d \dashv (-)^d$ —"— —"— $a \times d \to c$ and $b \times d \to c$

$d \times a \to c$ and $d \times b \to c$.

Again using $a \times d \cong d \times a$, $b \times d \cong d \times b$

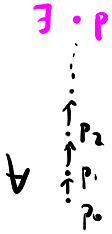
An arrow $d \times a \to c$ can be constructed as the composite: $(c^a \times c^b) \times a \xrightarrow{\pi_1 \times id_a} c^a \times a \xrightarrow{\varepsilon_{a,c}} c$

$d \times b \to c$ $(c^a \times c^b) \times b \xrightarrow{\pi_2 \times id_b} c^b \times b \xrightarrow{\varepsilon_{b,c}} c$

In short: we imitate logical/λ-calculus reasoning "inside" the category C.

Exercise 45. Let P be a poset. An *ω-chain* in P is a countable sequence $(p_i)_{i\in\omega}$ such that $p_i \leq p_{i+1}$ for every $i \in \omega$. The poset P is *ω-chain-complete* if any ω-chain has a supremum in P. A function $f\colon P \to Q$ between ω-chain-complete posets is a *morphism* if it preserves suprema of ω-chains. Prove that the category of ω-chain-complete posets is cartesian closed.

Remark. One may use this cartesian closed category to obtain a model of the untyped λ-calculus, by constructing a non-trivial ω-chain-complete U that is isomorphic to both $U \times U$ and U^U. This method was pioneered by D. Scott (1972). For more information, see, e.g., Ch. 18 in J. Lambek, P. J. Scott, *Introduction to higher order categorical logic*, Cambridge University Press (1986).



Given P, Q ω-cc we need to constract $P \times Q$ and Q^P. (Terminal object, $\mathbf{1}$, is the singleton)

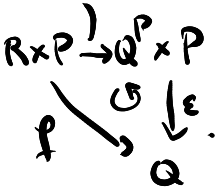
For $\underline{P \times Q}$, we use the product of posets. Check: if P, Q are ω-cc, then so is $P \times Q$.

For $\underline{Q^P}$, we take the set of morphisms $P \to Q$, and we equip it with a partial order:

$f \leq_{Q^P} g$ iff $f(p) \leq_Q g(p)$ for all $p \in P$; $\varepsilon_{P,Q} : Q^P \times P \to Q$, $(f, p) \mapsto f(p)$

We need to check several things:

- $(Q^P, \leq_{Q^P})$ is ω-chain complete.
- $\varepsilon_{P,Q}$ is a morphism.
- $\varepsilon_{P,Q}$ is universal: $\forall f : R \times P \to Q \;\; \exists! f' : R \to Q^P$ s.t.

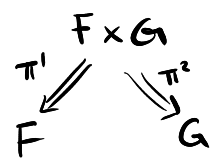


Exercise 46. Let C be a small category. Recall that $\widehat{C}$ is the category of functors $C^{\mathrm{op}} \to \mathbf{Set}$, with natural transformations between them. Prove that $\widehat{C}$ is cartesian closed.

Hint. First assume the exponential G^F exists, and use the Yoneda lemma to rewrite $G^F(a)$ as the set of natural transformations from $\mathbf{y}a$ to G^F.

$\widehat{C}$ has finite products: $(F \times G)(a) := Fa \times Ga$ defines a product $F \times G$ of $F, G \in \mathrm{ob}\,\widehat{C}$, with projections

$$(F\times G)\begin{pmatrix} a \\ \downarrow f \\ b \end{pmatrix} := \begin{matrix} Fa \times Ga \\ \uparrow Ff \times Gf \\ Fb \times Gb \end{matrix}$$



where $\pi^1_a : Fa \times Ga \to Fa$ is the 1st proj.
$\pi^2_a : Fa \times Ga \to Ga$ 2nd proj.
(in Set), for any $a \in \mathrm{ob}\,C$.

terminal object: $1 : C^{\mathrm{op}} \to \mathbf{Set}$ "constant functor at $\{*\}$"

$$a \mapsto \{*\}$$

$$\begin{matrix} a \\ f\downarrow \\ b \end{matrix} \mapsto \begin{matrix} \{*\} \\ \uparrow \mathrm{id} \\ \{*\} \end{matrix}$$

$\widehat{C}$ has exponentials: [if G^F exists, then by Yoneda, for any object a of C, we have $G^F(a) \cong \hat{C}(ya, G^F)$

Then, by adjunction, $\hat{C}(ya, G^F) \cong \hat{C}(ya \times F, G)$ since $(-)\times F \dashv (-)^F$.]

We define, for any $a \in \mathrm{ob}\,C$, $G^F(a) := \hat{C}(ya \times F, G)$.

for any $f : a \to b$ in C, $G^F(f) : \hat{C}(yb \times F, G) \longrightarrow \hat{C}(ya \times F, G)$.

$$(\beta : yb \times F \Rightarrow G) \longmapsto \alpha : ya \times F \Rightarrow G$$

$$\beta_c : yb c \times Fc \to Gc \qquad (\alpha_c : C(c,a) \times Fc \to Gc)_{c \in \mathrm{ob}\,C}$$

where $\alpha_c(g, x) := \beta_c(f \circ g, x)$ for any $(g, x) \in C(c,a) \times Fc$

To check: α is a nat. trans; G^F is a functor.

To show: G^F is exponential, we define $\varepsilon : G^F \times F \Rightarrow G$, with component at $a \in \mathrm{ob}\,C$,

$$\varepsilon_a : \hat{C}(ya \times F, G) \times Fa \to Ga$$

$$(\alpha, x) \longmapsto \alpha_a(\mathrm{id}_a, x).$$

To check: ε is nat. trans.

ε is universal.

$\alpha : ya \times F \Rightarrow G$

$\alpha_a : yaa \times Fa \to Ga$